

HALIDES

When metallic elements combine with halogen elements, halides are formed. The halogens are iodine, fluorine, chlorine, and bromine. Halide minerals are commonly very soft and of low specific gravity, often having crystals classified in the cubic system. Many of these minerals, such as halite and sylvite, form in evaporite sequences by the drying out of saline waters. Other halides – for example, fluorite – occur in hydrothermal veins.

FLUORITE

Cubic • H 4 • SG 3.18

This calcium fluoride often forms transparent to translucent crystals of various colours. Large quantities are used for making hydrofluoric acid.



YELLOW FLUORITE



PURPLE FLUORITE



cubic crystal

SYLVITE

Cubic • H 2 • SG 1.99

Sylvite is a potassium chloride, a salt similar to halite. It occurs with halite in evaporite deposits. Sylvite is used to make potash fertilizers.

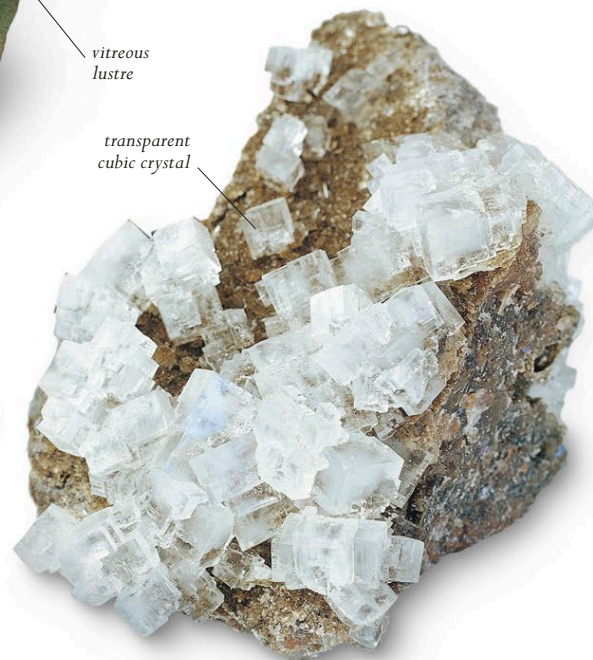


granular carnallite

CARNALLITE

Orthorhombic • H 2½ • SG 1.6

Carnallite is a hydrated chloride of magnesium and potassium, formed by the evaporation of saline water. It is important in fertilizer manufacture.



vitreous lustre

transparent cubic crystal

HALITE CRYSTALS



DIABOLITE

Tetragonal • H 2½ • SG 5.42

A lead copper chloride hydroxide with a light to dark blue colour, diabolite forms by the alteration of other minerals.



BOLEITE

Cubic • H 3–3½ • SG 5.05

Deep blue boleite is a rare hydroxide of lead, silver, copper, and chlorine. It occurs where lead and copper deposits have been altered.



ORANGE HALITE

HALITE

Cubic • H 2 • SG 2.17

Halite (common salt), a sodium chloride, occurs in extensive beds formed by the evaporation of saline water. It can be coloured or colourless.



GREEN FLUORITE